Session 4 Pre-Work

Zotero MCP and Your Workspace

Instructor: Brady Allardice — brady.allardice@upf.edu

Session 4 introduces **MCPs** — the way Claude Code talks to systems outside your project. You will use one of them in an exercise: **Zotero**, to write a short literature review from a practice library. It needs to be installed and connected before class — the setup is too fiddly to do in the room. (I will demo a GitHub MCP myself; you do not need to install it.)

Please complete **all four steps** below. Budget **30–45 minutes total**. If you get stuck for more than 10 minutes, stop and email me — don't try to debug it yourself. Once you are done, run the verification commands at the end and **send me a screenshot** of the output.

What You Need Before Starting

- Everything from the original Pre-Workshop Setup Guide working (Claude Code authenticated, VS Code running, Python installed)
- A free Zotero account — we'll create one in Step 3 if you don't have one

If `claude -version` or `git -version` doesn't work, fix that first — these steps assume the original setup is intact.

Step 1: Refresh the Course Repo and Set Up Your Workspace

Starting with Session 4 we are switching to a new workflow: the course repo stays **read-only** on your side, and you do all your actual work in a separate `~/my_workspace/` folder. This means I can push new material any time without breaking whatever you have in progress, and you can edit/break/delete freely without worrying about merge conflicts against the shared repo.

Run these three commands now, and plan to run them at the start of every future session too:

```
# 1.  Pull the latest course repo
cd ~/claude-code-workshop && git pull

# 2.  Copy today's session folder into your workspace
mkdir -p ~/my_workspace && cp -r session_4/ ~/my_workspace/session_4/

# 3.  Open your copy in VS Code and work there
cd ~/my_workspace/session_4 && code .
```

If you cloned the repo somewhere other than `~/claude-code-workshop` (e.g. inside Dropbox or Documents), substitute your actual path in step 1. If you don't already have the course repo cloned at all, run this first:

```
cd ~ && git clone https://github.com/bradyallardice/claude-code-workshop.git
```

After Step 2, you should see a `session_4/` folder inside `~/my_workspace/` containing the new material. After Step 3, VS Code should open in that copy. From now on, **edit files in `~/my_workspace/session_4/`, not in the course repo itself.**

If `git pull` fails with a merge conflict, that is exactly the problem we fix in class — don't try to resolve it. Just email me and I will help you reset cleanly.

Step 2: Update Claude Code

The MCP install commands below assume Claude Code 2.1 or newer. Update to the latest version:

```
claude update
```

Then confirm the version:

```
claude --version
```

Expected: 2.1.x or higher.

Step 3: Install the Zotero MCP

The Zotero MCP lets Claude Code search your Zotero library, pull BibTeX, and insert citations into a draft. I demo it in class, then you use it in the mini literature review exercise, and for your own papers afterward.

3a. Set up Zotero and import the practice library

- If you don't have a Zotero account, create a free one at **https://www.zotero.org/user/register**
- Install the **Zotero desktop app** from **https://www.zotero.org/download/** if you don't already have it — the MCP needs the desktop app running to read your library (including full-text PDFs)
- Open the desktop app and sign in

Now import the practice library. I have prepared a bundle of 9 journal articles (with 10 PDF attachments — one article has two PDFs) for the in-class demo and exercise. It lives in the course repo at:

```
session_4/docs/claude_code_zotero/
```

After running `git pull` from Step 1, that folder contains a `claude_code_zotero.rdf` file and a `files/` subfolder of PDFs.

To import:

- In the Zotero desktop app, click **File** → **Import...**
- Choose **"A file (BibTeX, RIS, Zotero RDF, etc.)"**, click **Continue**
- Navigate to `session_4/docs/claude_code_zotero/` and select `claude_code_zotero.rdf`
- On the next screen, check **"Place imported collections and items into new collection"** and **"Copy files to the Zotero storage folder"**
- Click **Continue**

You should end up with a new collection called `claude_code_zotero` in the left sidebar containing 9 papers (top-level items), with 10 PDF attachments in total — one paper has two PDFs. **Confirm by clicking on the collection and verifying you see 9 papers, each with an attachment paperclip icon, and that expanding them shows 10 PDFs in total.**

Critical: keep the `files/` folder *next to* the `.rdf` on disk during import — if you move just the .rdf or import it from elsewhere, Zotero will import the metadata but not the PDFs.

3b. Let Claude Code talk to your Zotero app

Zotero blocks other applications from connecting by default. Turn that off:

- Open the Zotero desktop app
- **Mac:** **Zotero → Settings** **Windows:** **Edit → Settings**
- Go to the **Advanced** tab
- Under **General**, check the box **"Allow other applications on this computer to communicate with Zotero"**
- Close the Settings window. **Leave Zotero running** — the MCP needs it open to query your library

The MCP also falls back to the Zotero Web API if the desktop app is closed, but the local connection is faster and gives Claude Code access to your PDFs — worth turning on.

3c. Get your Zotero API key and Library ID

- Go to **https://www.zotero.org/settings/keys**
- Your **Library ID** (a number, e.g. `1234567`) is shown at the top under "Your userID for use in API calls"
- Click **Create new private key**
- **Key name:** `claude-code-workshop`
- Check **Allow library access** (read-only is fine; check write access too if you want Claude to add items)
- Click **Save Key**, then **copy the key immediately**

Save both the key and your Library ID into a temporary scratch file.

3d. Install the `zotero-mcp-server` package

Mac:

```
pip install zotero-mcp-server
```

If `pip` is not found, use:

```
pip3 install zotero-mcp-server
```

Windows:

```
py -m pip install zotero-mcp-server
```

Make sure you type `zotero-mcp-server` (not `zotero-mcp`) in the install command: a different, unrelated package on PyPI is called `zotero-mcp`. The package we use installs a command-line program called `zotero-mcp`.

Then confirm it installed:

```
zotero-mcp --help
```

Expected: a help message listing options. If you see "command not found," close and reopen your terminal first.

3e. Add the Zotero MCP to Claude Code

Run (substituting your real values from Step 3c):

```
claude mcp add --scope user zotero -e ZOTERO_LOCAL=true -e ZOTERO_LIBRARY_ID=YOUR_LIBRARY_ID -e
ZOTERO_API_KEY=YOUR_API_KEY -e ZOTERO_LIBRARY_TYPE=user - zotero-mcp serve
```

Then verify:

```
claude mcp list
```

You should now see `zotero` listed with ✓ `Connected`. **Delete the scratch file with your key and Library ID.**

`ZOTERO_LOCAL=true` tells the MCP to read from the Zotero desktop app (the setting you turned on in Step 3b), which is what gives Claude Code access to your PDFs. The API key and Library ID are used for the Zotero web API, which the MCP falls back to if the desktop app is not running. **The web API only sees the practice collection once the Zotero desktop app has finished syncing it to your online library**, so after the import, leave Zotero open until the sync icon at the top right stops spinning. `--scope user` is what makes the server available in every folder.

3f. Confirm it works

Open Claude Code in any directory and ask:

```
Use the Zotero MCP to list 5 items from the claude_code_zotero collection.
```

If it returns titles from the practice library (Alt et al., Bisgaard, Hicks et al., etc.), you are done. **Make sure the Zotero desktop app is open** when you run this — if it isn't, the MCP will either fail or fall back to the (slower) Web API.

Step 4: Re-Read the Session 3 Git Cycle (5 min)

Session 4's sharing-with-coauthors material and the optional capstone build on the Git workflow from Session 3. Spend five minutes re-reading the slides from Session 3 (`slides/session3_slides.pdf` in the course repo), specifically the `commit` → `instruct` → `diff` → `decide` cycle and how branches work.

Verification: Confirm Everything Works

Open a terminal and run each of the following. **Take a screenshot of the full output and send it to me**.

```
claude --version
```

Expected: `2.1.x` or higher.

```
claude mcp list
```

Expected: `zotero` listed with ✓ `Connected`.

```
zotero-mcp --help
```

Expected: a help message (any output is fine; "command not found" is the failure mode).

Also confirm:

- `git pull` on the course repo succeeds and you see a `session_4/` folder with new content
- The `claude_code_zotero` collection in your Zotero app contains 9 papers with 10 PDF attachments in total (one paper has two PDFs)
- Asking Claude Code "list 5 items from the claude_code_zotero collection" returns titles

Troubleshooting

`claude mcp add` fails with "unknown flag": You are on an older Claude Code. Run `claude update` (Step 2) and try again.

Zotero MCP shows "Connected" but returns empty results: Your library is genuinely empty, or `ZOTERO_LIBRARY_TYPE` is wrong. If you set up a *group* library instead of a personal one, change `user` to `group` in the env var.

`zotero-mcp: command not found` after `pip install`: Close and reopen your terminal. If still failing, the install went to a directory not on your PATH — email me with the output of `pip show zotero-mcp-server` and I will help.

`claude mcp list` shows `zotero` in one folder but not in another: The server was registered with the default "local" scope, which only applies to the folder where you ran `claude mcp add`. Re-run the command with `--scope user` (as shown above). If Claude Code says the server already exists, first remove the old entry with `claude mcp remove zotero` and then run the `claude mcp add` command again.

You accidentally pasted your Zotero key into a chat or committed it to git: Revoke it immediately (**https://www.zotero.org/settings/keys**) and generate a new one. Tokens are easy to rotate; a leaked token in a public repo is not.

Anything else: Email me at brady.allardice@upf.edu. Do not spend more than 10 minutes troubleshooting on your own. I would much rather help you before the workshop than lose time during it.